

Kubernetes (K8s) vs Amazon EKS

Beginner-Friendly DevOps Guide with Real-World Examples

Goal: Understand what Kubernetes is, what EKS is, how they are different, and how a DevOps engineer uses them in real projects.

1. First: What Problem Does Kubernetes Solve?

Suppose you have a web application packaged as Docker containers. You may need 10, 50, or 500 containers running across multiple servers. You also need to restart failed containers, distribute traffic, scale applications, perform rolling deployments, and keep the desired number of replicas running.

Kubernetes automates these container-management tasks.

Example: Your application should always have 3 copies running. If one Pod crashes, Kubernetes detects the failure and creates another Pod so the desired state returns to 3.

Simple flow: **Developer** → **Docker Image** → **Kubernetes** → **Pods** → **Application**

2. What is Kubernetes (K8s)?

Kubernetes is an **open-source container orchestration platform**. It manages containers across a cluster of machines.

The name Kubernetes is often shortened to **K8s**: K + 8 letters + s.

Main responsibilities: deployment, scaling, service discovery, load balancing, self-healing, rolling updates, configuration, secrets, and workload scheduling.

3. Important Kubernetes Components

Component	Easy meaning	Example
Cluster	Complete Kubernetes environment	Production K8s environment
Control Plane	Brain of the cluster	Decides where Pods should run
Node	Machine that runs workloads	EC2/VM/physical server
Pod	Smallest deployable K8s unit	Pod containing an Nginx container
Deployment	Maintains desired Pod replicas	Keep 3 web Pods running
Service	Stable network endpoint	Expose web Pods internally

4. Kubernetes Architecture — Simple View

Control Plane

API Server → Scheduler → Controller Manager → etcd

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Worker Nodes

Node 1 → Pod A, Pod B

Node 2 → Pod C, Pod D

Node 3 → Pod E

Easy analogy: Control Plane = manager/brain. Worker Nodes = workers. Pods = application units.

5. What is Amazon EKS?

Amazon EKS (Elastic Kubernetes Service) is AWS's managed Kubernetes service.

EKS gives you a Kubernetes cluster integrated with AWS. AWS manages the Kubernetes **control plane**, while you run your workloads using supported compute options such as managed node groups, self-managed nodes, or AWS Fargate.

Key idea: EKS is not a different container orchestration technology. It is **Kubernetes provided as an AWS managed service**.

6. K8s vs EKS — The Most Important Difference

Area	Self-managed Kubernetes	Amazon EKS
Technology	Kubernetes	Kubernetes
Control Plane	You operate it	AWS operates the EKS control plane
AWS Integration	You configure it	Deep AWS integration
Worker Compute	VMs/physical servers/etc.	EC2, managed node groups, Fargate, etc.
Operations	More responsibility	AWS reduces control-plane operational work
Best for	Learning, custom environments, on-prem/cloud-native dev	AWS-focused production workloads

7. Real-World Example — Company Uses Kubernetes

Scenario: An e-commerce company has a Java application with a frontend, backend API, and payment service.

Without Kubernetes: DevOps engineers manually start containers, monitor failures, configure networking, and scale instances.

With Kubernetes: Deployments define the desired number of replicas; Services provide stable networking; Kubernetes reschedules failed Pods.

Example: Backend deployment requests 3 replicas. One Pod crashes → Kubernetes creates a replacement → traffic continues through the Service.

8. Real-World Example — Same Application on EKS

Scenario: The same company already uses AWS services such as VPC, IAM, EC2, CloudWatch, and Load Balancer.

They create an EKS cluster. AWS operates the EKS control plane. The DevOps team deploys Kubernetes objects such as Deployments and Services. Worker workloads run on EC2-based nodes or other supported EKS compute options.

Typical flow:

Developer → Git → Jenkins/CI → Docker Image → ECR → EKS → Pods → AWS Load Balancer → Customers

9. Very Simple Analogy

Kubernetes = Restaurant management system

You install it, configure it, operate it, and maintain the management infrastructure.

EKS = AWS-managed restaurant management system

AWS takes care of the Kubernetes control-plane operation, while your team focuses more on applications, configurations, security, networking, and worker compute.

Remember: **EKS does not replace Kubernetes. EKS provides managed Kubernetes.**

10. DevOps Commands You Should Know

Check cluster connection:

```
kubectl get nodes
```

List Pods:

```
kubectl get pods
```

List Pods in all namespaces:

```
kubectl get pods -A
```

Describe a Pod:

```
kubectl describe pod <pod-name>
```

Apply a Kubernetes manifest:

```
kubectl apply -f deployment.yaml
```

Check EKS clusters using AWS CLI:

```
aws eks list-clusters --region ap-south-1
```

Update local kubeconfig for an EKS cluster:

```
aws eks update-kubeconfig --region ap-south-1 --name my-eks-cluster
```

11. Practical Deployment Example

Step 1: Build the application into a Docker image.

Step 2: Push the image to a container registry such as Amazon ECR.

Step 3: Create an EKS cluster and worker compute.

Step 4: Connect kubectl to the cluster.

Step 5: Deploy the application using a Kubernetes Deployment.

Step 6: Create a Service to provide stable networking.

Step 7: Expose the application using an AWS-integrated load-balancing solution where required.

Step 8: Monitor logs, metrics, health, scaling, and deployment status.

12. Interview Questions & Answers

Q1. What is Kubernetes?

Kubernetes is an open-source container orchestration platform used to deploy, scale, manage, and heal containerized applications.

Q2. What is EKS?

EKS is Amazon's managed Kubernetes service. AWS manages the Kubernetes control plane, while customers manage their workloads and selected infrastructure.

Q3. Is EKS different from Kubernetes?

EKS is a managed service based on Kubernetes. Kubernetes is the underlying open-source orchestration platform.

Q4. Why use EKS instead of self-managed Kubernetes?

For AWS workloads, EKS can reduce control-plane operational effort and provides strong integration with AWS networking, identity, compute, storage, and observability services.

Q5. Who manages the EKS control plane?

AWS manages the EKS control plane.

Q6. What is a Pod?

A Pod is the smallest deployable unit in Kubernetes and can contain one or more closely related containers.

Q7. What is a Node?

A Node is a machine that provides compute capacity for Kubernetes workloads.

13. One-Line Memory Trick

K8s = Kubernetes technology

EKS = AWS-managed Kubernetes service

Interview-ready sentence: "Kubernetes is an open-source container orchestration platform, whereas Amazon EKS is AWS's managed Kubernetes service where AWS operates the Kubernetes control plane."

14. What to Learn Next

For a DevOps Engineer, learn in this order:

1. Docker → 2. Kubernetes basics → 3. Pods/Deployments/Services → 4. ConfigMaps & Secrets → 5. Ingress → 6. Volumes & Storage → 7. Probes → 8. RBAC → 9. Helm → 10. EKS → 11. AWS IAM/VPC/Load Balancers → 12. CI/CD with Jenkins → 13. Monitoring and troubleshooting.